Attribute	Details
University / Institute	L J University — [IMCA]
Semester / Division	Sem-3 | Div-D
Subject	UDP
Group No.	Group-14
Back End Designer	Ayush Vyas (D-54)
Front End Designer	Rohit Pareek (D-58)
Database Designer	Het Moradiya (D-24)

Sec No.	Title	Page No.
1.	Introduction	04
1.1	Existing System	04
1.2	Need for the New System	04
1.3	Objective of the New System	04
1.4	Problem Definition	05
1.5	Core Components	05
1.6	Scope of Project	05
1.7	Assumptions and Constraints	06
1.8	Advantages and Limitations	06
2.	Project Planning	07

2.1	Team Details and Roles	07
3.	Requirement Determination & Analysis	08
3.1	Requirement Determination	08
3.2	Targeted Users	08
4.	System Design	08
4.1	Use Case Diagram	09
4.2	Data Dictionary	09
5.	Proposed Enhancements	14
6.	Conclusion	15
7.	Bibliography & References	16

DOCUMENTATION
PROJECT : HOTEL ELITE
DEFINATION: HOTEL MANAGEMENT SYSTEM

1. INTRODUCTION

1.1 Existing System

Manual Registers: All guest details, room bookings, and payments are written by hand in paper notebooks or diaries.

Risk of Errors: High chance of human mistakes, such as booking the same room for two different guests at the same time

Time-Consuming: Check-in and check-out processes take a lot of time because staff must fill out everything manually on paper.

Communication Gap: The front desk has to call or physically walk to the housekeeping staff to check if a room is clean or dirty.

1.2 Need for the New System

Centralized Data: To manage all hotel activities like bookings, bills, staff logs, and kitchen orders from a single computer screen.

Fast & Paperless: To save time and storage space by storing records digitally instead of using paper registers.

Real-Time Tracking: To see the live status of every room instantly on a screen.

Online Booking: To allow customers to view room photos, check pricing, and book their rooms from anywhere in the world.

1.3 Objective of the New System

Automation: To automate room pricing based on seasons or weekends and calculate the final bill instantly.

Role-Based Security: To protect data by creating password-secured pages for Admin, Receptionist, and Housekeeping. No user can view pages outside their role.

Business Analytics: To show visual sales and profit graphs to the hotel owner to help them understand daily, weekly, and monthly growth.

1.4 Problem Definition

"The traditional manual system used in hotels causes booking mistakes, money calculation errors, and poor communication among staff. This makes customers unhappy and reduces hotel revenue. To solve these problems permanently, we are creating 'Hotel Elite', a fast, secure, and fully digital web application."

1.5 Core Components

Room Management: Manages room types and tracks their live status .

Booking & Reservation: Displays an easy calendar view to manage both online website bookings and walk-in customers.

Guest Management: Stores guest information safely, including names, phone numbers, ID cards, and past visit history.

Billing & Invoicing: Combines room prices, food bills, and taxes into one final, professional receipt.

Food & Room Service: Customers scan a QR code inside the room with their phone, browse the menu, and order food straight to the kitchen.

Housekeeping: Shows a clean-up checklist for the staff. Once they finish cleaning, they change the status back to 'Available'.

Reports & Analytics: Creates simple, colorful graphs for the Admin to track hotel income and customer feedback.

1.6 Project Scope :

In-Scope:Room booking, customized dashboards for Admin/Receptionist/Housekeeping, QR code scanning from rooms to order food, and automatic bill generation.

Out-of-Scope : It does not connect to real bank payment systems . It is designed for only one hotel, not for multiple hotel chains.

1.7 Assumptions and Constraints

Assumptions: The hotel will always have a stable internet connection so that staff devices can stay connected.

Constraints: The system needs internet to function. It is designed to fit within the short timeline of the college semester.

1.8 Advantages and Limitations

Advantages: Zero double-booking mistakes, lower paper costs, faster operational speed, and 100% accurate financial calculations.

Limitations: Staff members require basic training to use a computer or smartphone. The application stops working during total network outages.

2.PROJECT PLANNING

2.1 Team Details and Roles

Back End Designer - [D54 Ayush Vyas]:

Role: Responsible for backend development, server logic, and connecting the application with the database.

Front End Designer - [D58 Rohit Pareek]:

Role: Responsible for designing user interfaces, website layouts, buttons, and screen designs.

Database Designer - [D24 Het Moradiya]:

Role: Responsible for creating database schemas, managing data tables, and defining data relationships.

3. REQUIREMENT DETERMINATION & ANALYSIS

3.1 Requirement Determination

Functional Requirements: Secure login system, room status updates, bill generation, QR code food ordering, and report charts.

Non-Functional Requirements: Fast speed , high security , and a fully responsive screen design for mobiles and computers.

3.2 Targeted Users

Admin : Has full access. Manages staff accounts, updates room rates, and checks business profit reports.

Receptionist: Has medium access. Checks room availability, handles customer check-in/check-out, and prints final bills.

Housekeeping Staff: Has limited access. Reviews the dirty room list and updates status to 'Available' once cleaning is finished.

Customer: Has public access. Browses room pictures, checks available dates, books a room, and orders food inside the hotel.

4. SYSTEM DESIGN

4.1 Use Case Diagram

Customer: Browse Rooms, Book Rooms, Order Food.

Receptionist: Check-In Guests, Print Invoice, Check Available Rooms.

Housekeeping: View Dirty Rooms, Mark Rooms as Clean.

Admin: View Profit Analytics, Create Staff Accounts, plus all features above.

4.2 Data Dictionary

Table 1: admin

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique admin identification number
username	VARCHAR(50)	NOT NULL	Admin username for secure login
password	VARCHAR(255)	NOT NULL	Encrypted/Plain admin password

Table 2: users

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique user identification number
name	VARCHAR(100)	NOT NULL	Full name of the registered user
email	VARCHAR(255)	UNIQUE, NOT NULL	User email address for login
password	VARCHAR(255)	NOT NULL	Secure hashed password (bcrypt)

Table 3: rooms

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique room ID
room_no	VARCHAR(20)	NOT NULL	Room number identifier (e.g., 101, 105)

Column Name	Data Type	Constraint	Description
floor	VARCHAR(50)	NULLABLE	Building floor location (e.g., 1st Floor)
room_type	VARCHAR(50)	NULLABLE	Category (Single, Deluxe, Executive Suite, etc.)
room_status	VARCHAR(50)	Default: 'Clean & Ready'	Current cleanliness / readiness status
price	DECIMAL(10,2)	NOT NULL (Default: 1800.00)	Base price per night stay
bed_type	VARCHAR(100)	NULLABLE	Bed configuration (Queen Size, King Size Luxury, etc.)
image	VARCHAR(255)	NULLABLE	Image URL / reference for room display
max_adults	INT(11)	Default: 2	Maximum adult occupancy allowed
max_children	INT(11)	Default: 1	Maximum children occupancy allowed
extra_adult_charge	DECIMAL(10,2)	Default: 500.00	Additional charge per extra adult
extra_child_charge	DECIMAL(10,2)	Default: 300.00	Additional charge per extra child

Table 4: room_bookings

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique booking record ID
phone	VARCHAR(15)	NULLABLE	Guest contact phone number
id_proof_type	VARCHAR(100)	NULLABLE	Type of identification document (PAN, Passport, etc.)
id_proof_number	VARCHAR(100)	NULLABLE	Identification document number
booking_date	TIMESTAMP	Default: CURRENT_TIMESTAMP	Timestamp when booking was created
room_id	INT(11)	NULLABLE	Foreign key reference to rooms table
room_no	VARCHAR(50)	NULLABLE	Room number at time of booking
check_in	DATETIME	NULLABLE	Scheduled check-in date and time

Column Name	Data Type	Constraint	Description
check_out	DATETIME	NULLABLE	Scheduled check-out date and time
user_email	VARCHAR(255)	NULLABLE	Guest email address
user_name	VARCHAR(255)	NULLABLE	Guest full name
total_price	DECIMAL(10,2)	NULLABLE	Total calculated room rent amount
payment_method	VARCHAR(50)	Default: 'Cash'	Payment mode (Card, Cash, Online)
payment_status	VARCHAR(50)	Default: 'Pending'	Payment status (Paid Successfully, Pending)

Table 5: food_items

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique food item ID
food_name	VARCHAR(255)	NOT NULL	Name of the dish or beverage
price	DECIMAL(10,2)	NOT NULL	Price of the food item
image	VARCHAR(255)	NULLABLE	Image URL of the food item

Table 6: food_orders

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique food order ID
customer_name	VARCHAR(100)	NULLABLE	Name of ordering guest
room_no	VARCHAR(20)	NULLABLE	Room number for delivery
items	TEXT	NULLABLE	List of ordered food items and prices
total_amount	INT(11)	NULLABLE	Total bill amount for the food order
payment_mode	VARCHAR(50)	NULLABLE	Mode of payment (Online, Cash)
order_date	TIMESTAMP	Default: CURRENT_TIMESTAMP	Time when order was placed
status	VARCHAR(50)	Default: 'Ordered'	Kitchen preparation status (Ordered, Preparing, Delivered)
payment_status	VARCHAR(50)	Default: 'Pending'	Food payment status (Paid, Pending)

Table 7: contact_messages

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique message ID
name	VARCHAR(100)	NOT NULL	Name of visitor sending message
email	VARCHAR(100)	NOT NULL	Email address of visitor
message	TEXT	NOT NULL	Content of inquiry or feedback
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Submission timestamp

Table 8: room_reviews

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique review ID
room_id	INT(11)	NOT NULL	Reference to reviewed room
guest_name	VARCHAR(100)	NOT NULL	Name of guest leaving review
rating	INT(11)	NOT NULL	Star rating score (e.g., 1 to 5)
comment	TEXT	NOT NULL	Feedback comment text
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Review submission timestamp

Table 9: staff

Column Name	Data Type	Constraint	Description
id	INT(11)	Primary Key, Auto Increment	Unique staff ID
name	VARCHAR(100)	NOT NULL	Full name of staff member
email	VARCHAR(100)	UNIQUE, NOT NULL	Staff login email
password	VARCHAR(255)	NOT NULL	Staff login password
role	VARCHAR(50)	Default: 'Staff'	Designated role (Manager, Kitchen Manager, Receptionist, Room Service, Housekeeping Lead)
work_assigned	TEXT	NULLABLE	Detailed responsibilities and

Column Name	Data Type	Constraint	Description
			duties description
status	VARCHAR(50)	Default: 'Active'	Employment / activity status
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Record creation timestamp

5. PROPOSED ENHANCEMENTS

AI Chatbot: An automated helper tool on the website to answer customer questions about pricing and room features immediately.

IoT Smart Room Keys: A digital key code sent directly to the customer's phone to open room doors without a physical metal key.

6. CONCLUSION

The Hotel Elite web application replaces slow, error-prone manual hotel records with a single, synchronized platform. It blocks double-booking errors, saves staff time, protects customer information, and provides a clear financial breakdown to the hotel owner.

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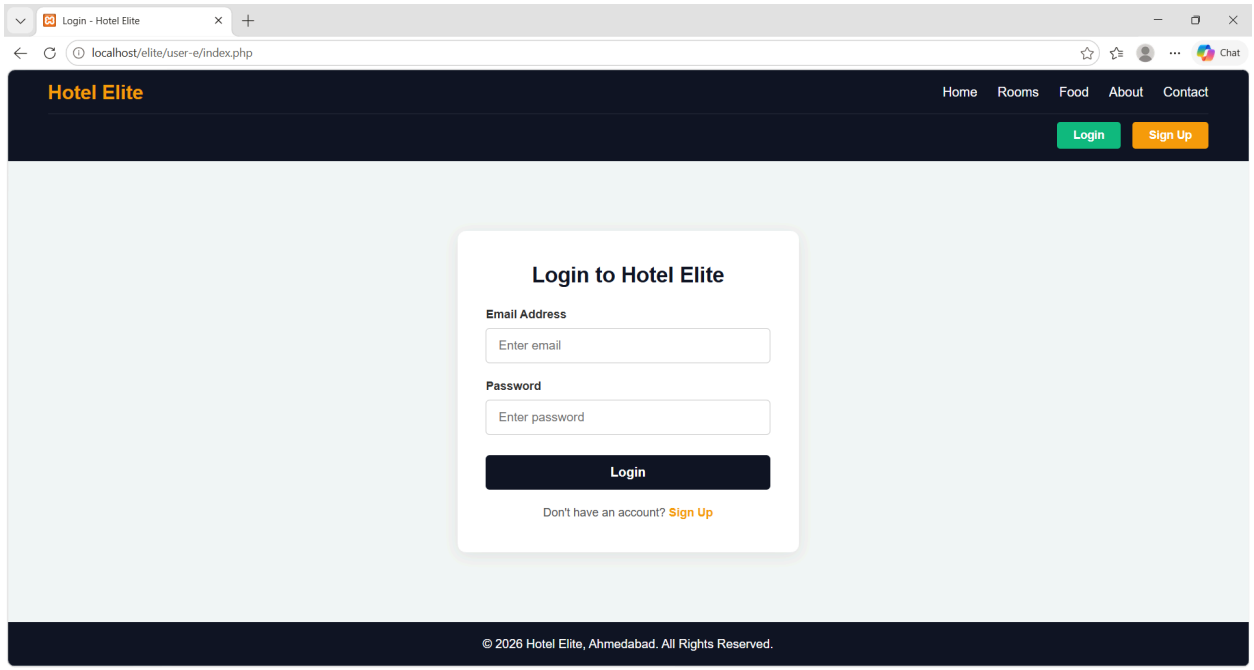
References: Prof.Pankaj sir

Institution:L J university

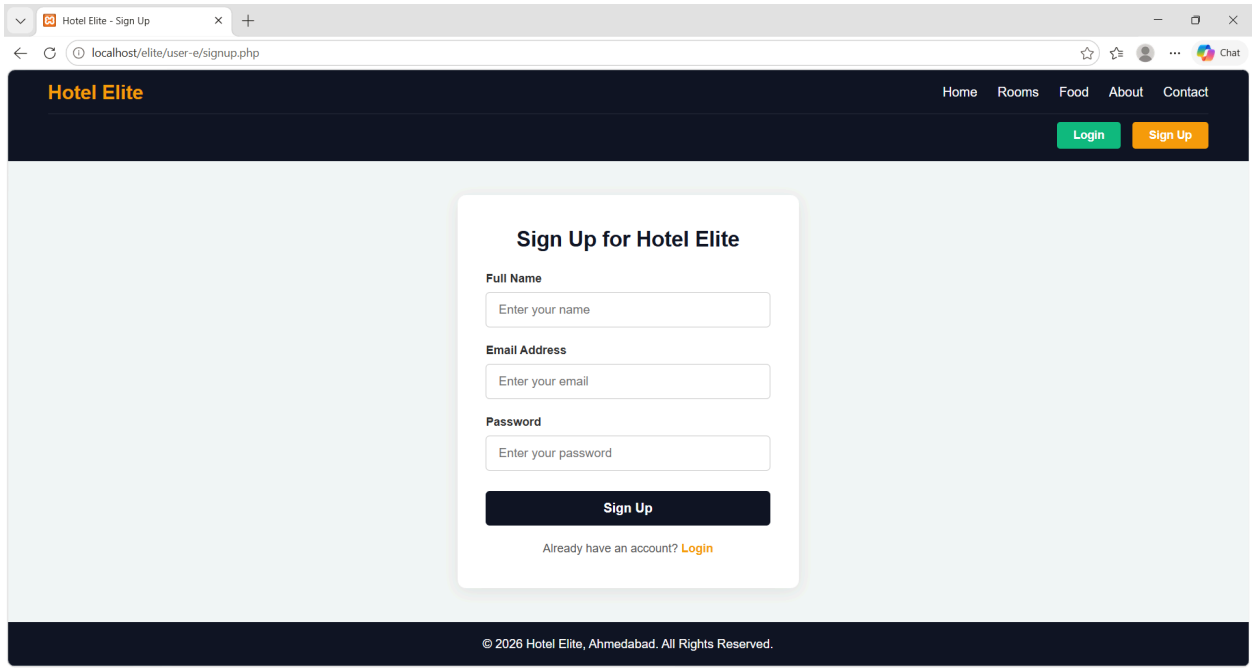
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USER PANEL

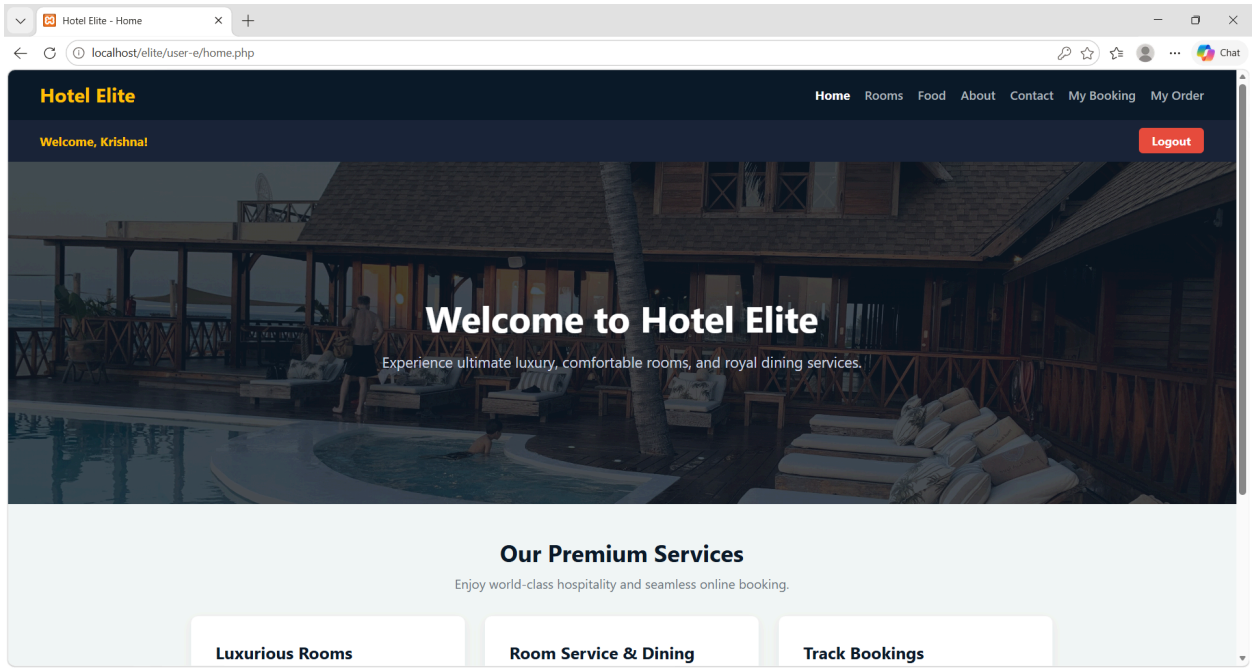
LOGIN PAGE

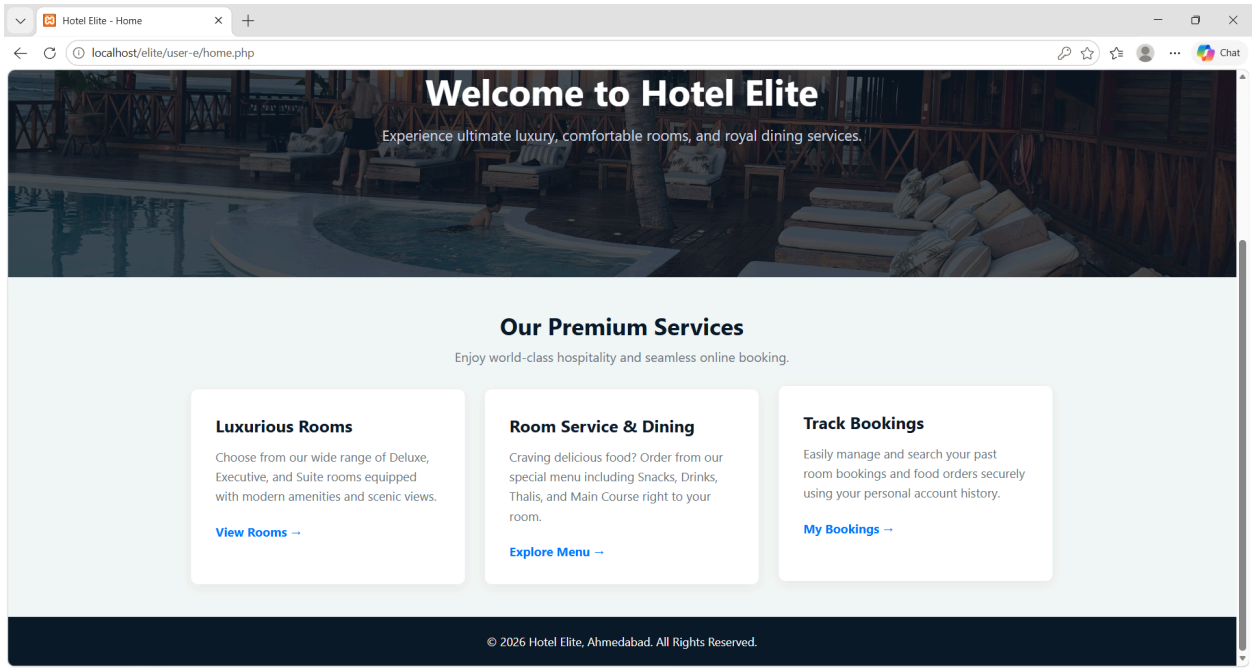


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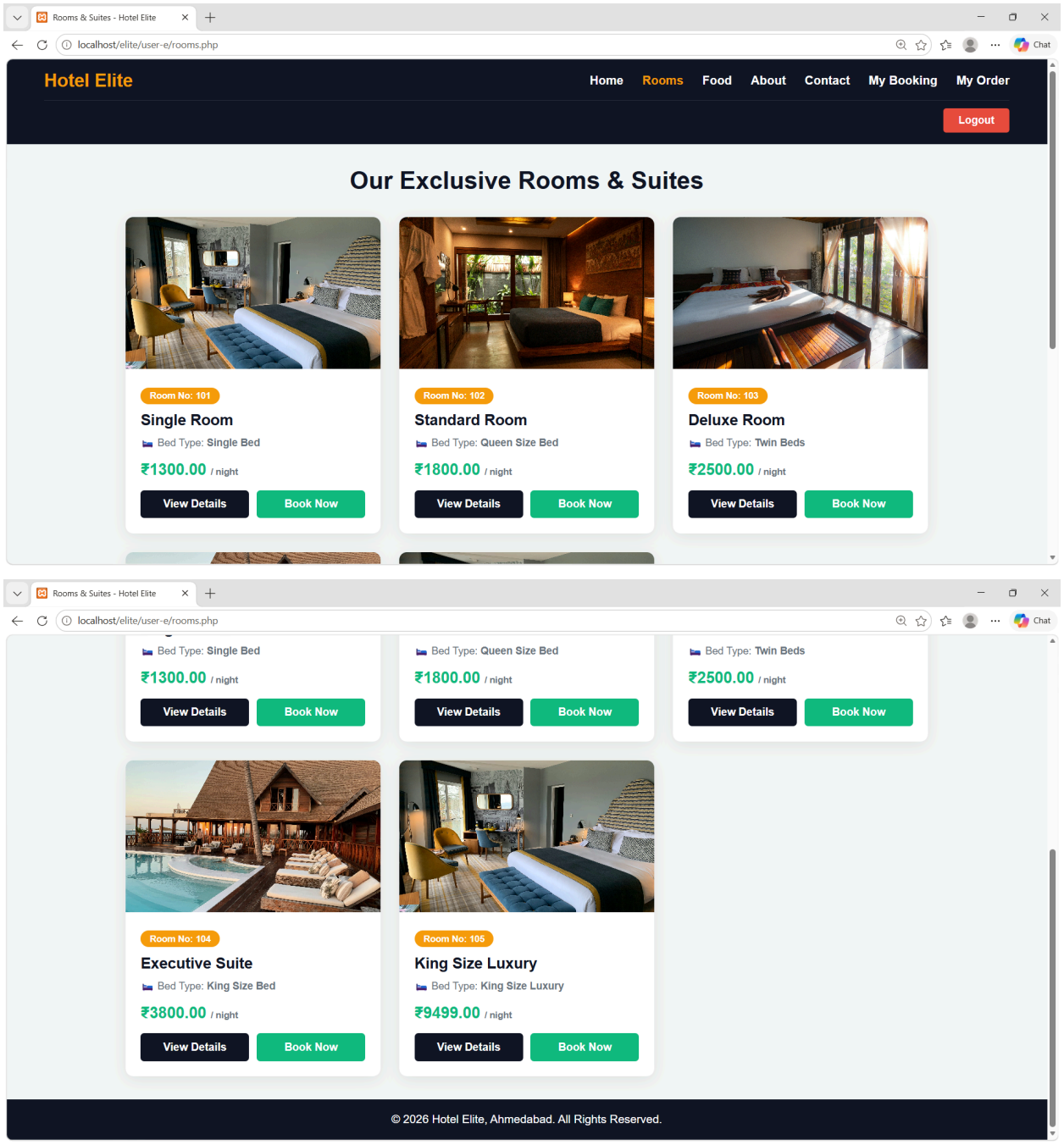


HOME PAGE

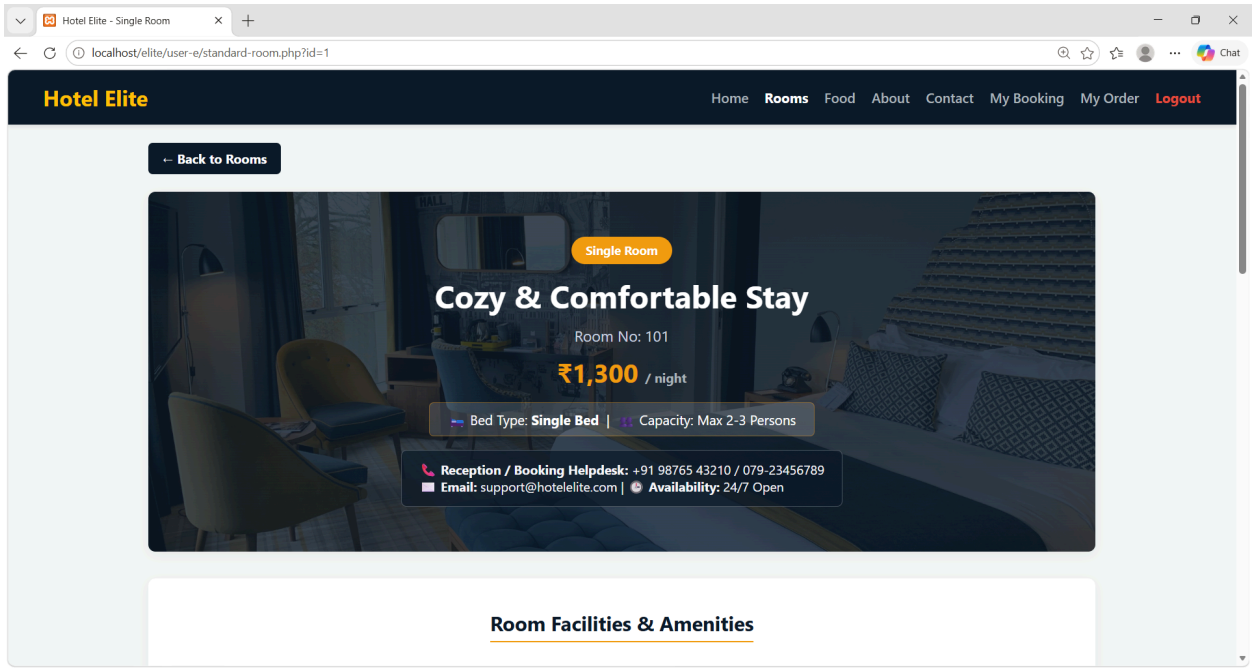

Hotel Elite - Home
localhost/elite/user-e/home.php
Chat
Hotel Elite
Home Rooms Food About Contact My Booking My Order
Welcome, Krishna!
Logout
Welcome to Hotel Elite
Experience ultimate luxury, comfortable rooms, and royal dining services.
Our Premium Services
Enjoy world-class hospitality and seamless online booking.
Luxurious Rooms
Room Service & Dining
Track Bookings

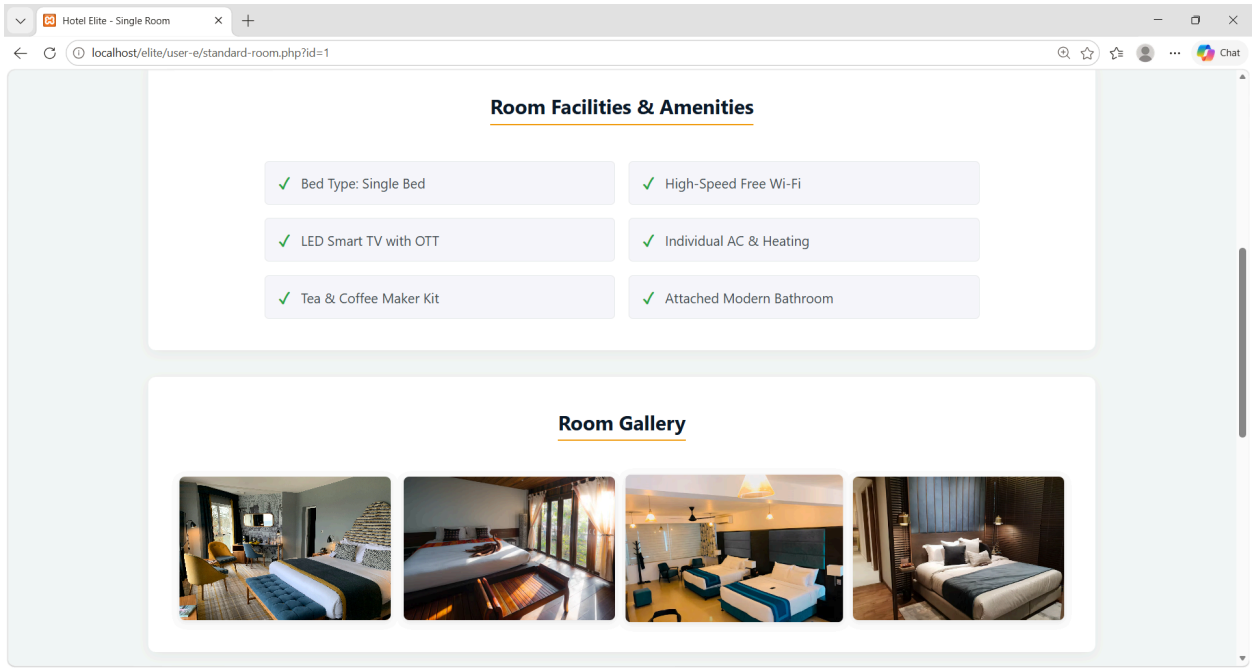

Hotel Elite - Home
localhost/elite/user-e/home.php
Chat
Welcome to Hotel Elite
Experience ultimate luxury, comfortable rooms, and royal dining services.
Our Premium Services
Enjoy world-class hospitality and seamless online booking.
Luxurious Rooms
Choose from our wide range of Deluxe, Executive, and Suite rooms equipped with modern amenities and scenic views.
View Rooms →
Room Service & Dining
Craving delicious food? Order from our special menu including Snacks, Drinks, Thalis, and Main Course right to your room.
Explore Menu →
Track Bookings
Easily manage and search your past room bookings and food orders securely using your personal account history.
My Bookings →
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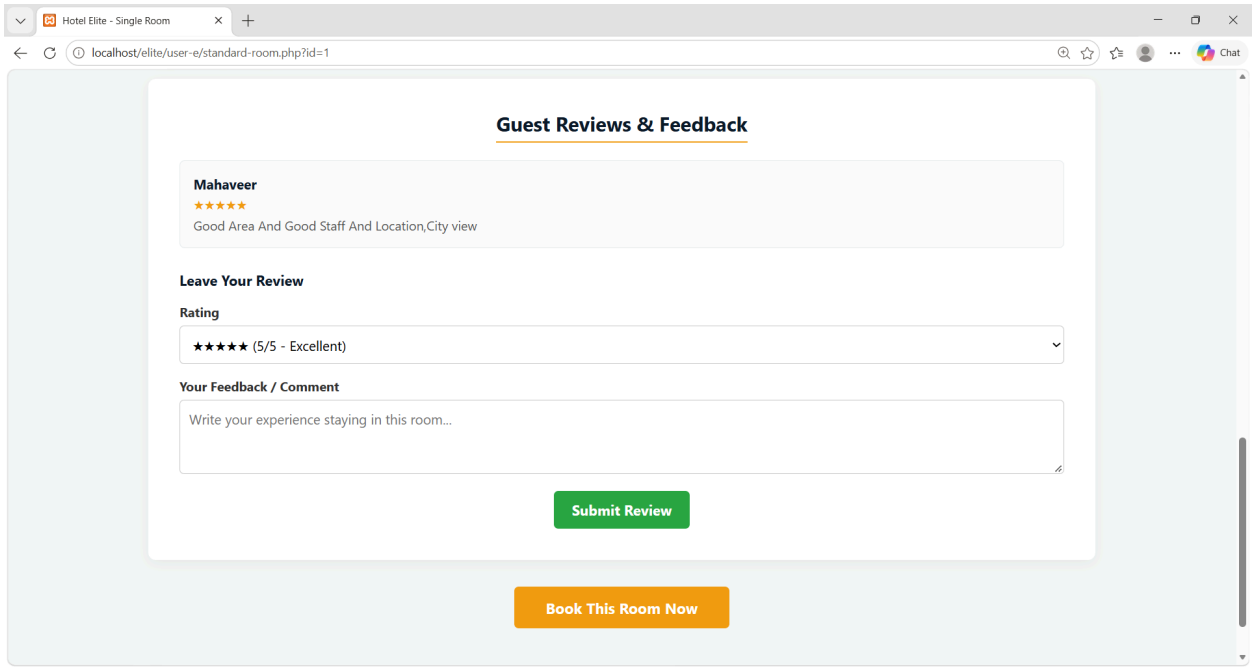
ROOMS PAGE


Rooms & Suites - Hotel Elite
localhost/elite/user-e/rooms.php
Chat
Hotel Elite
Home
Rooms
Food
About
Contact
My Booking
My Order
Logout
Our Exclusive Rooms & Suites
Room No: 101
Single Room
Bed Type: Single Bed
₹1300.00 / night
View Details
Book Now
Room No: 102
Standard Room
Bed Type: Queen Size Bed
₹1800.00 / night
View Details
Book Now
Room No: 103
Deluxe Room
Bed Type: Twin Beds
₹2500.00 / night
View Details
Book Now
Room No: 104
Executive Suite
Bed Type: King Size Bed
₹3800.00 / night
View Details
Book Now
Room No: 105
King Size Luxury
Bed Type: King Size Luxury
₹9499.00 / night
View Details
Book Now
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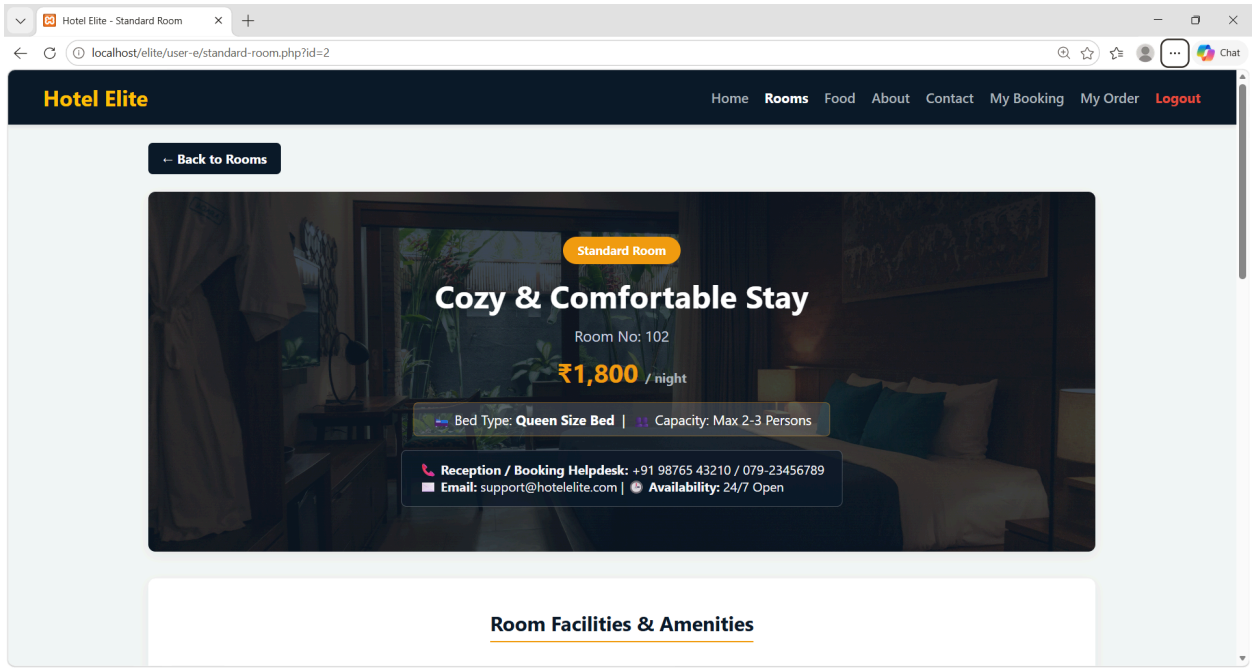
SINGLE ROOM


Hotel Elite - Single Room
localhost/elite/user-e/standard-room.php?id=1
Chat
Hotel Elite
Home
Rooms
Food
About
Contact
My Booking
My Order
Logout
← Back to Rooms
Single Room
Cozy & Comfortable Stay
Room No: 101
₹1,300 / night
Bed Type: Single Bed | Capacity: Max 2-3 Persons
Reception / Booking Helpdesk: +91 98765 43210 / 079-23456789
Email: support@hotelelite.com | Availability: 24/7 Open
Room Facilities & Amenities


Hotel Elite - Single Room
localhost/elite/user-e/standard-room.php?id=1
Chat
Room Facilities & Amenities
Bed Type: Single Bed
High-Speed Free Wi-Fi
LED Smart TV with OTT
Individual AC & Heating
Tea & Coffee Maker Kit
Attached Modern Bathroom
Room Gallery



STANDARD ROOM



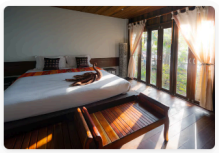
Hotel Elite - Standard Room

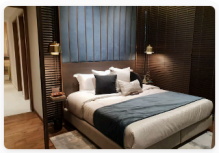
localhost/elite/user-e/standard-room.php?id=2

Room Facilities & Amenities

- ✓ Bed Type: Queen Size Bed
- ✓ High-Speed Free Wi-Fi
- ✓ LED Smart TV with OTT
- ✓ Individual AC & Heating
- ✓ Tea & Coffee Maker Kit
- ✓ Attached Modern Bathroom

Room Gallery



Hotel Elite - Standard Room

localhost/elite/user-e/standard-room.php?id=2

Guest Reviews & Feedback

No reviews for this room yet. Be the first guest to share your review below!

Leave Your Review

Rating

★★★★★ (5/5 - Excellent)

Your Feedback / Comment

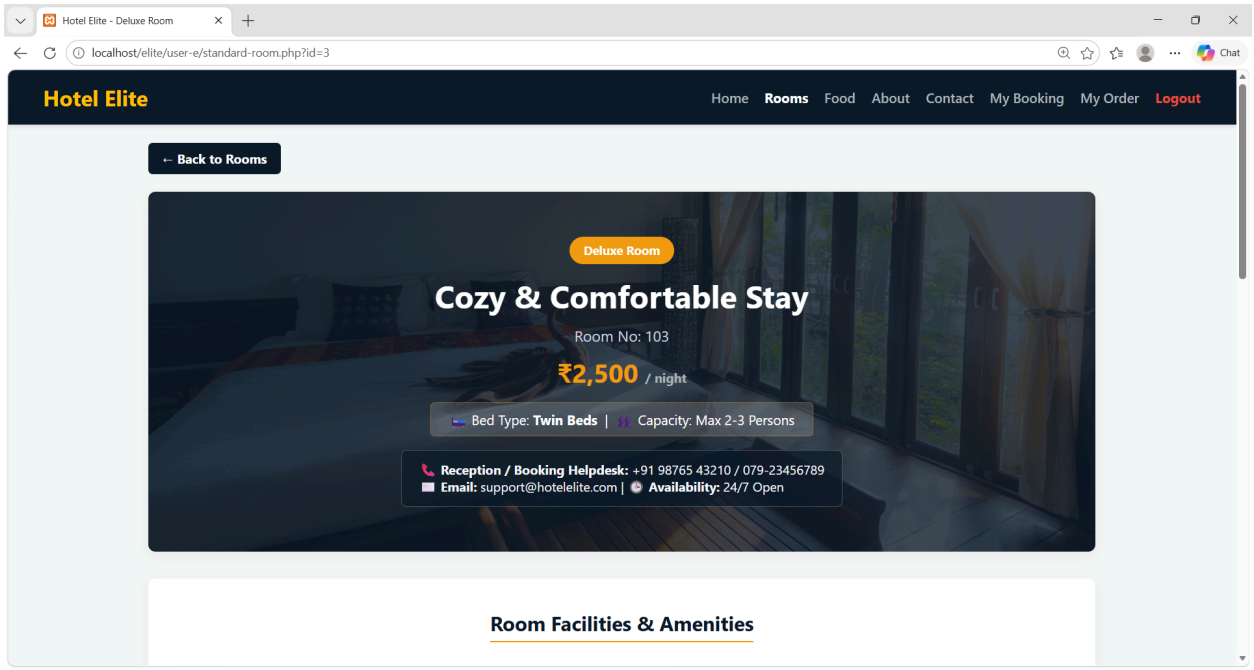
Write your experience staying in this room...

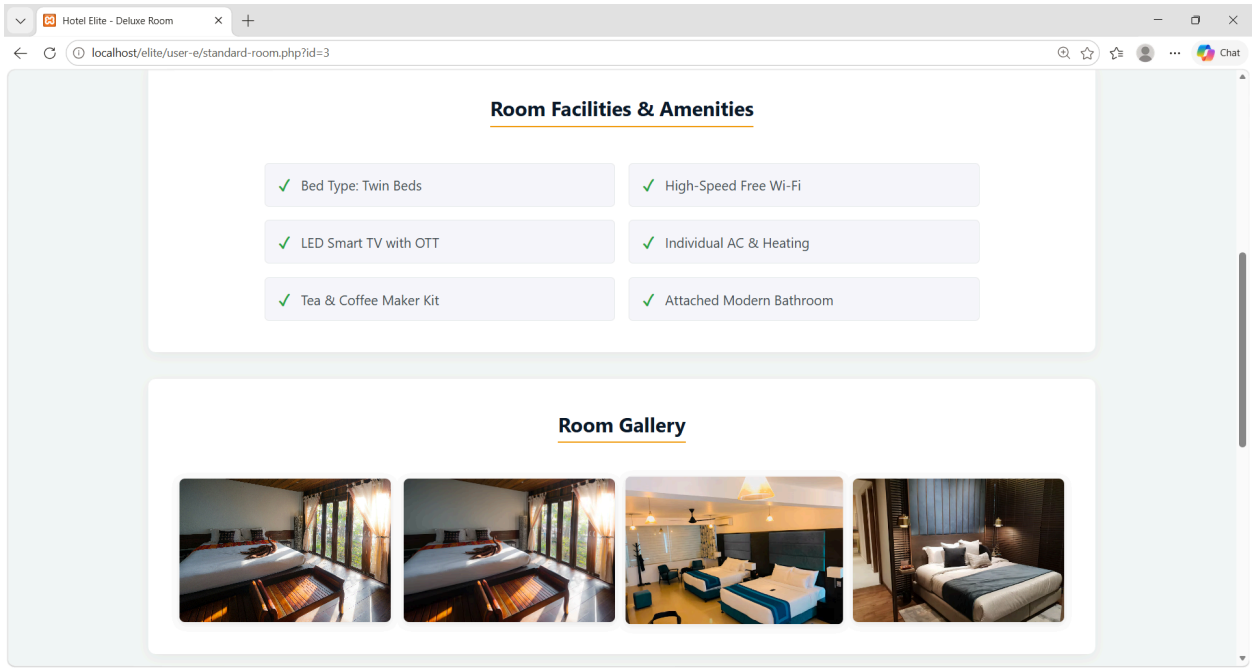
Submit Review

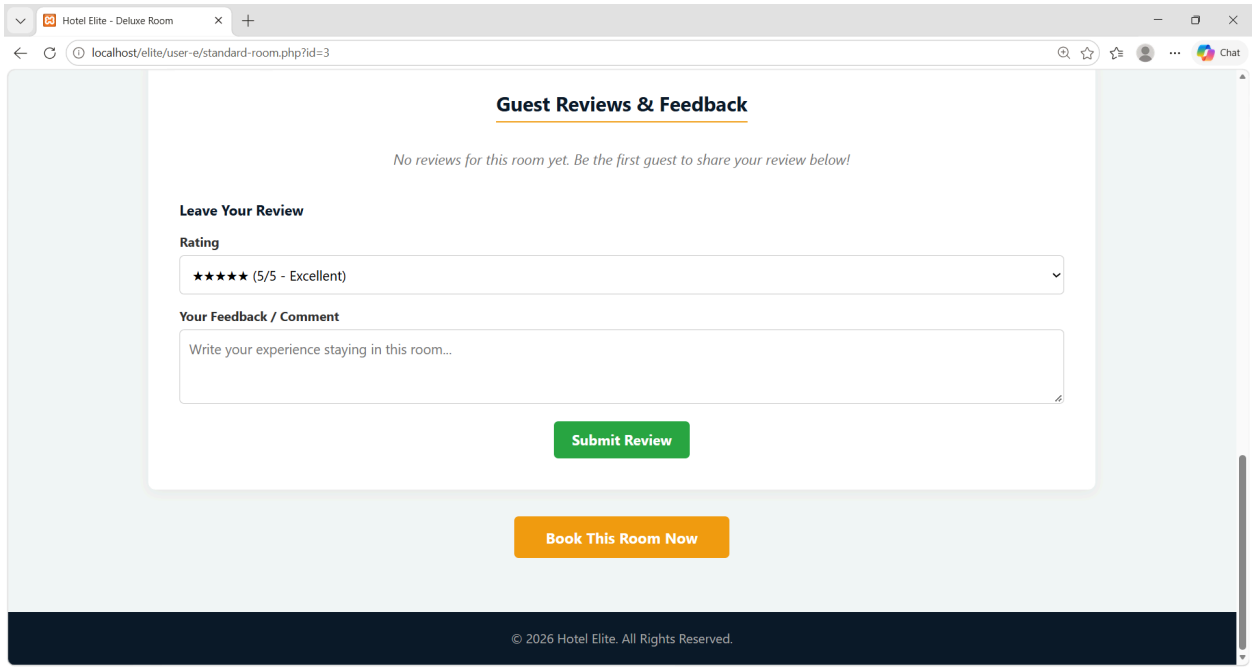
Book This Room Now

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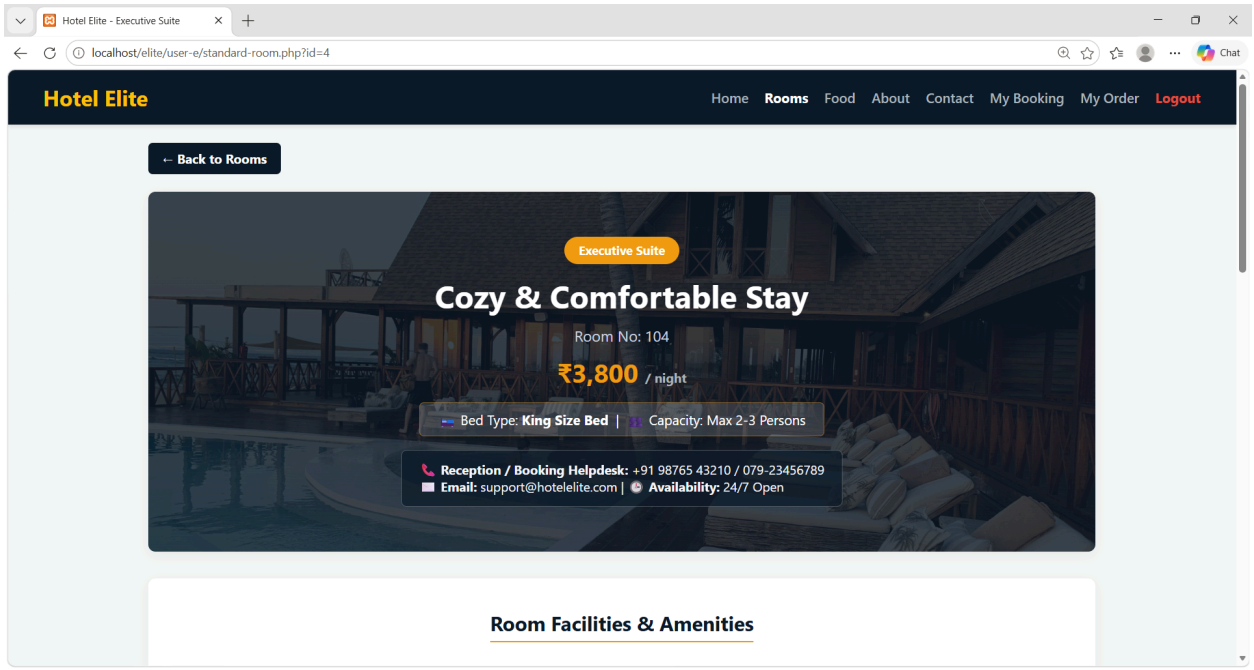
DELUXE ROOM


Hotel Elite - Deluxe Room
localhost/elite/user-e/standard-room.php?id=3
Chat
Hotel Elite
Home Rooms Food About Contact My Booking My Order Logout
← Back to Rooms
Deluxe Room
Cozy & Comfortable Stay
Room No: 103
₹2,500 / night
Bed Type: Twin Beds | Capacity: Max 2-3 Persons
Reception / Booking Helpdesk: +91 98765 43210 / 079-23456789
Email: support@hotelelite.com | Availability: 24/7 Open
Room Facilities & Amenities


Hotel Elite - Deluxe Room
localhost/elite/user-e/standard-room.php?id=3
Chat
Room Facilities & Amenities
✓ Bed Type: Twin Beds
✓ High-Speed Free Wi-Fi
✓ LED Smart TV with OTT
✓ Individual AC & Heating
✓ Tea & Coffee Maker Kit
✓ Attached Modern Bathroom
Room Gallery



EXECUTIVE SUITE



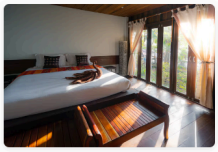
Hotel Elite - Executive Suite

localhost/elite/user-e/standard-room.php?id=4

Room Facilities & Amenities

- ✓ Bed Type: King Size Bed
- ✓ High-Speed Free Wi-Fi
- ✓ LED Smart TV with OTT
- ✓ Individual AC & Heating
- ✓ Private Jacuzzi & Bathtub
- ✓ Mini-Bar & Refrigerator
- ✓ 24/7 Personal Butler Service

Room Gallery



Hotel Elite - Executive Suite

localhost/elite/user-e/standard-room.php?id=4

Guest Reviews & Feedback

No reviews for this room yet. Be the first guest to share your review below!

Leave Your Review

Rating

★★★★★ (5/5 - Excellent)

Your Feedback / Comment

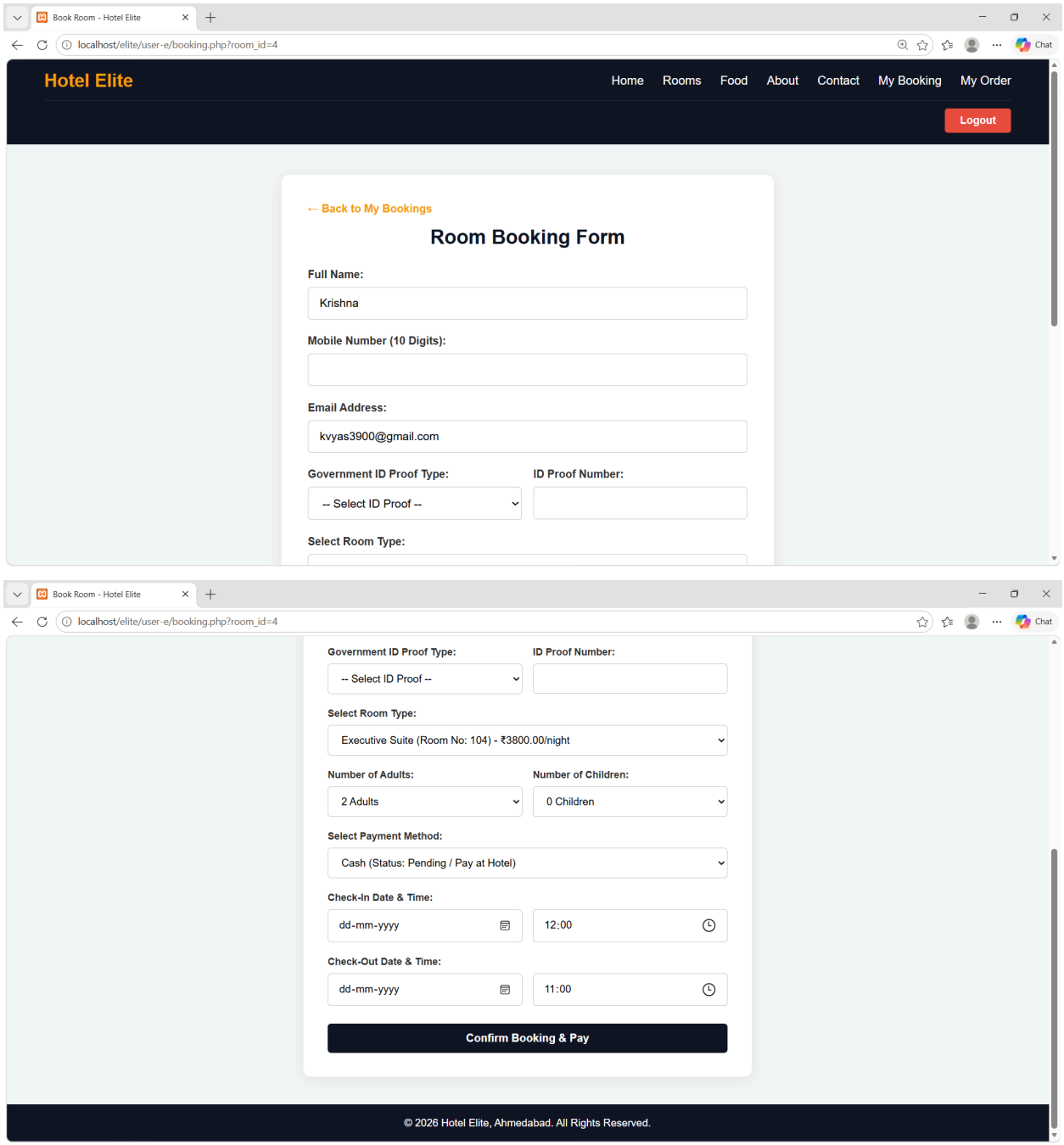
Write your experience staying in this room...

Submit Review

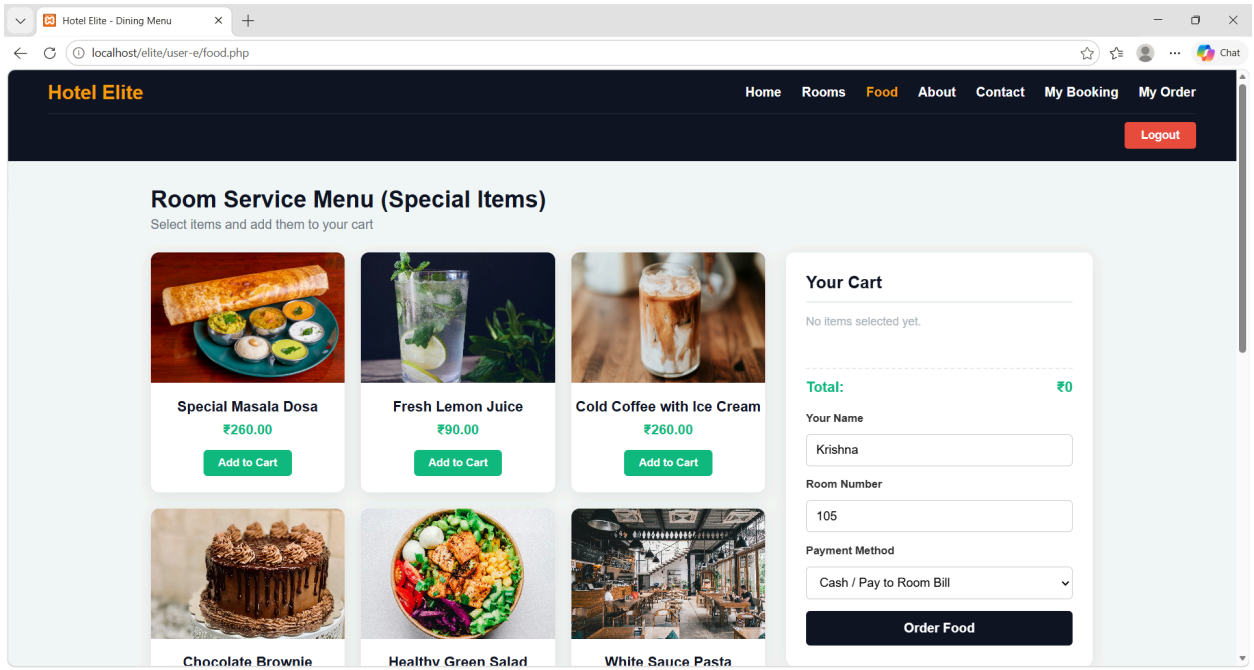
Book This Room Now

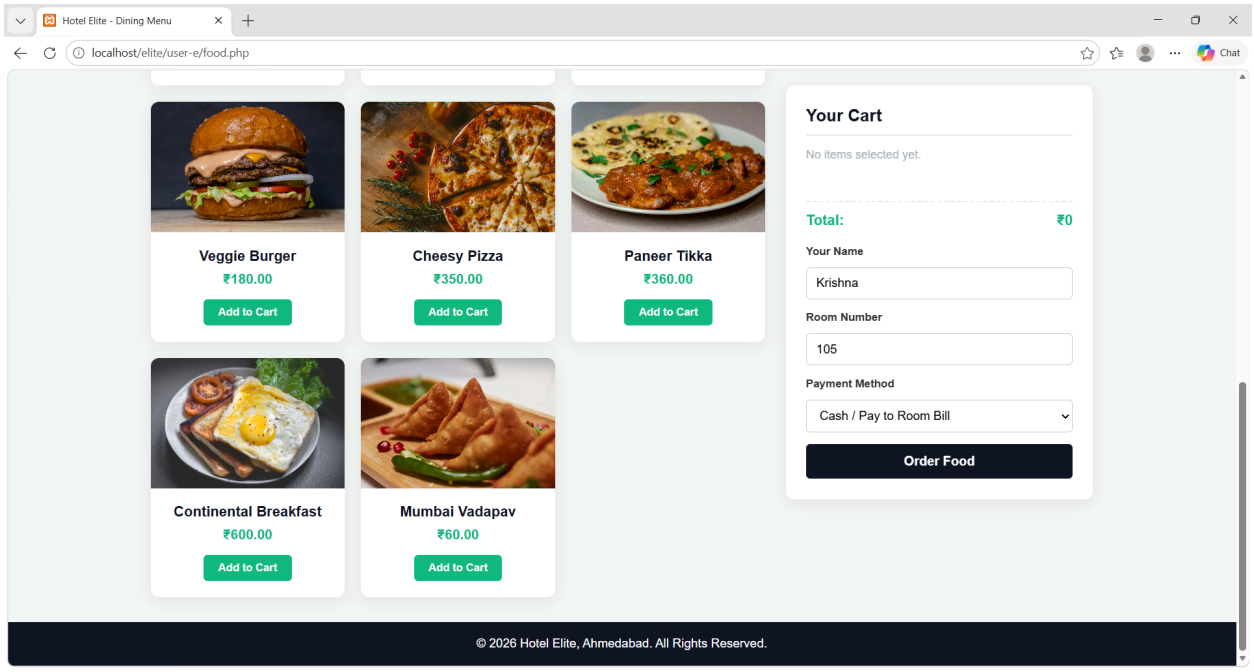
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BOOK THIS ROOM

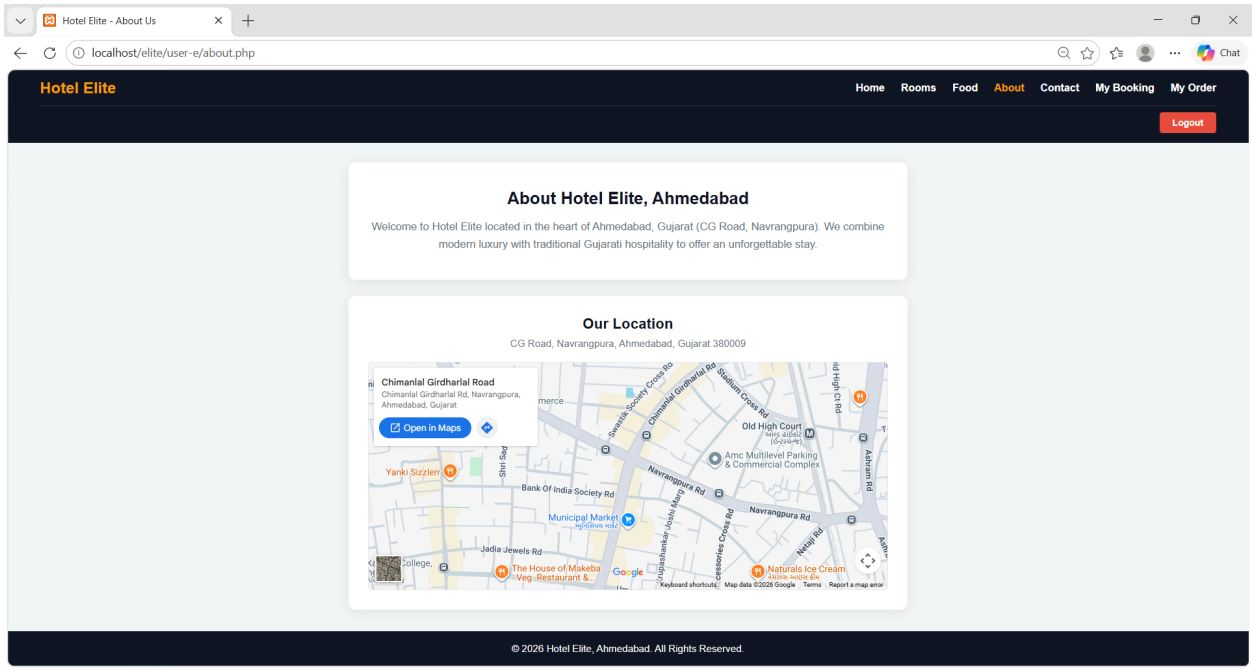


FOOD

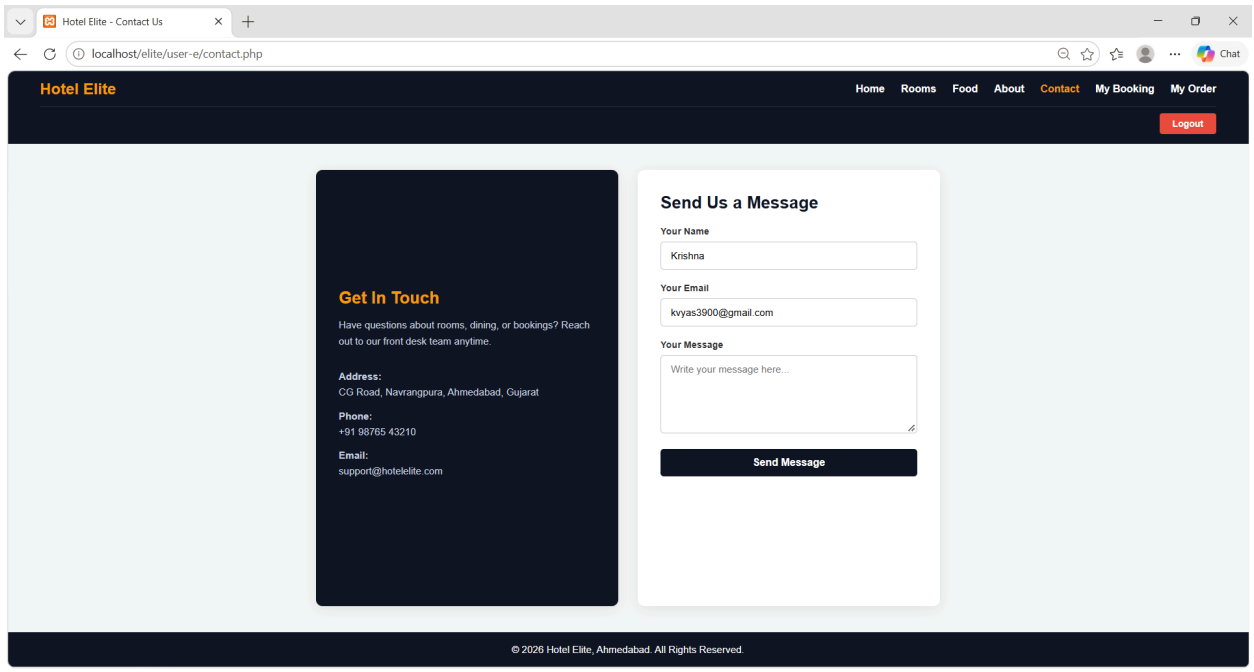

Hotel Elite - Dining Menu
localhost/elite/user-e/food.php
Chat
Hotel Elite
Home
Rooms
Food
About
Contact
My Booking
My Order
Logout
Room Service Menu (Special Items)
Select items and add them to your cart
Special Masala Dosa
₹260.00
Add to Cart
Fresh Lemon Juice
₹90.00
Add to Cart
Cold Coffee with Ice Cream
₹260.00
Add to Cart
Chocolate Brownie
Healthy Green Salad
White Sauce Pasta
Your Cart
No items selected yet.
Total:
₹0
Your Name
Krishna
Room Number
105
Payment Method
Cash / Pay to Room Bill
Order Food


Hotel Elite - Dining Menu
localhost/elite/user-e/food.php
Chat
Veggie Burger
₹180.00
Add to Cart
Cheesy Pizza
₹350.00
Add to Cart
Paneer Tikka
₹360.00
Add to Cart
Continental Breakfast
₹600.00
Add to Cart
Mumbai Vadapav
₹60.00
Add to Cart
Your Cart
No items selected yet.
Total:
₹0
Your Name
Krishna
Room Number
105
Payment Method
Cash / Pay to Room Bill
Order Food
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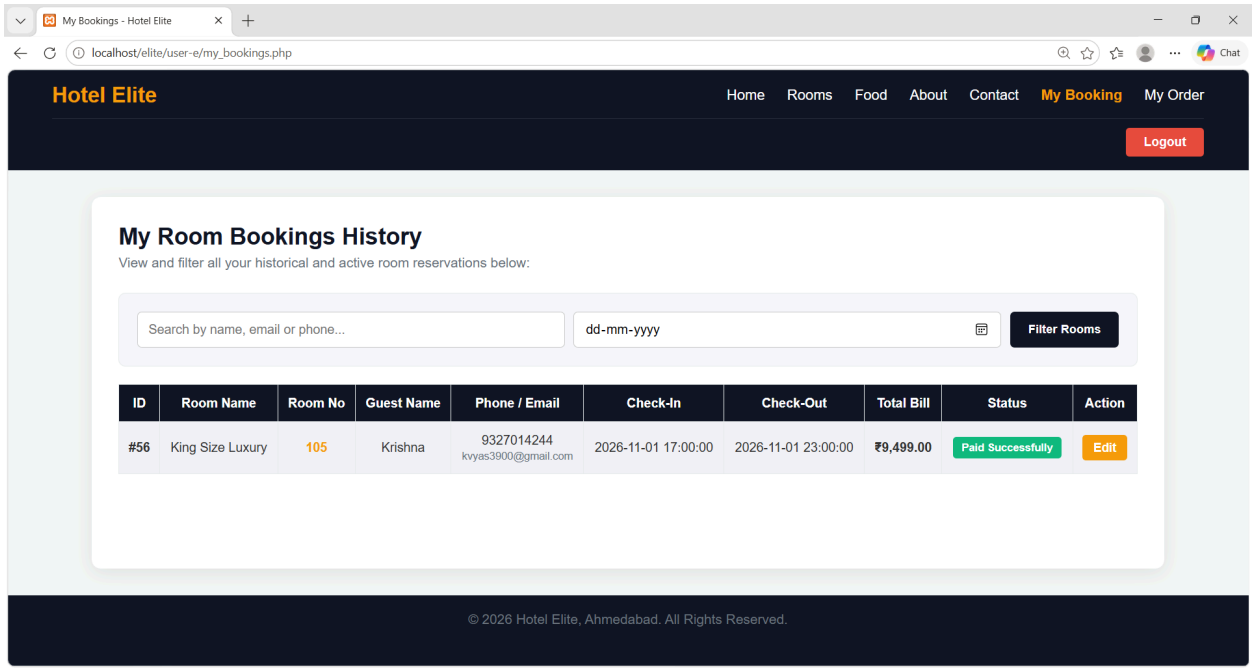
ABOUT



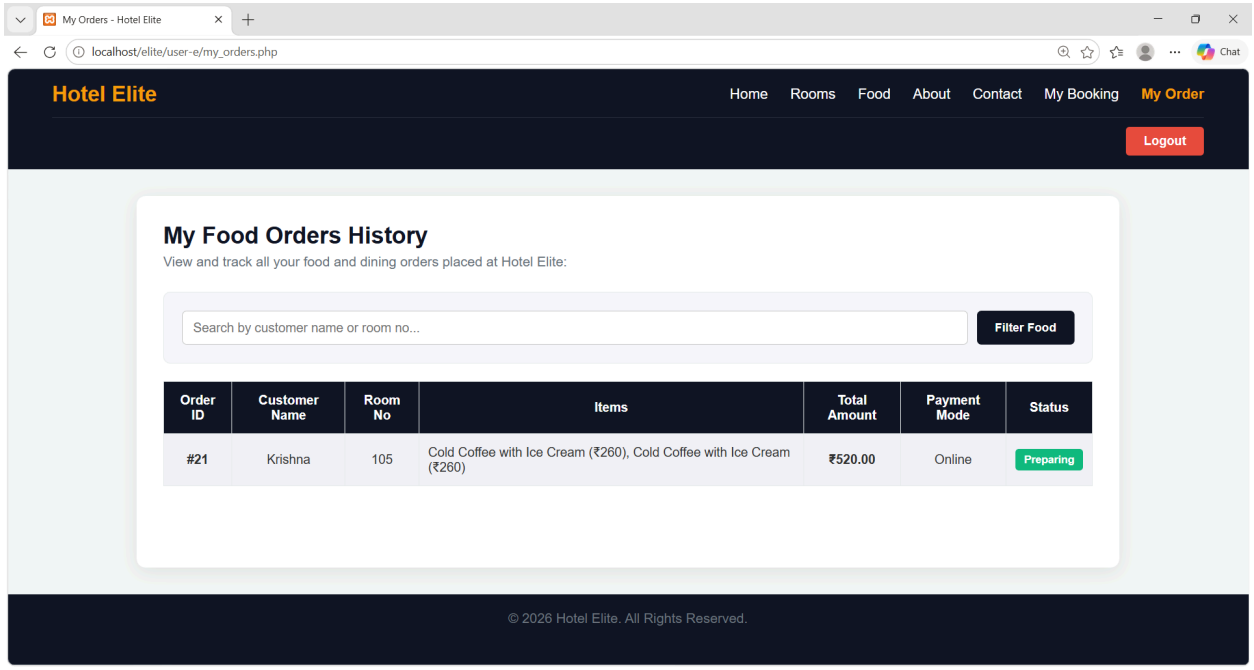
CONTACT



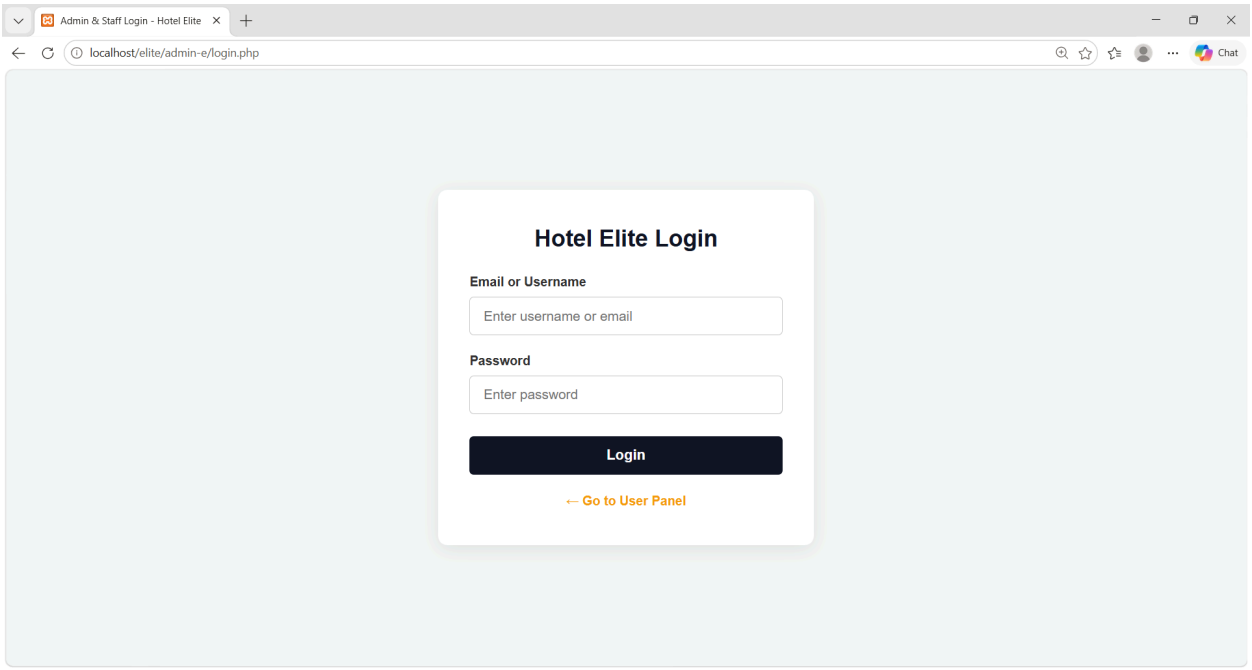
MY BOOKING



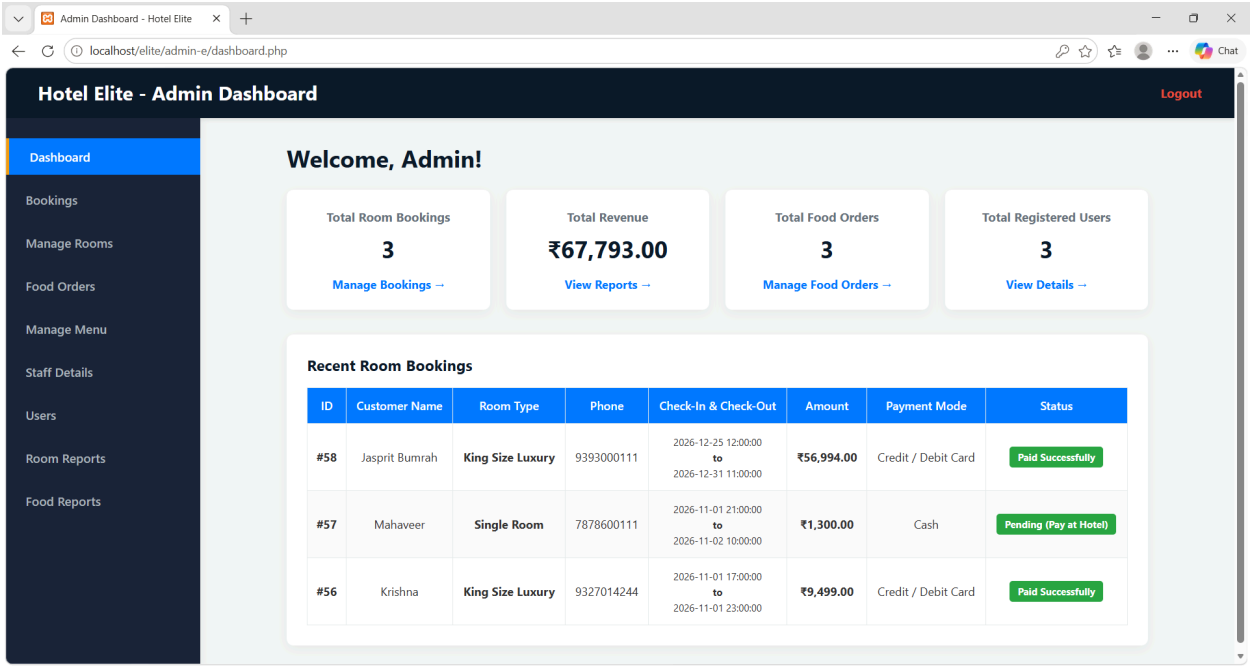
MY ORDER



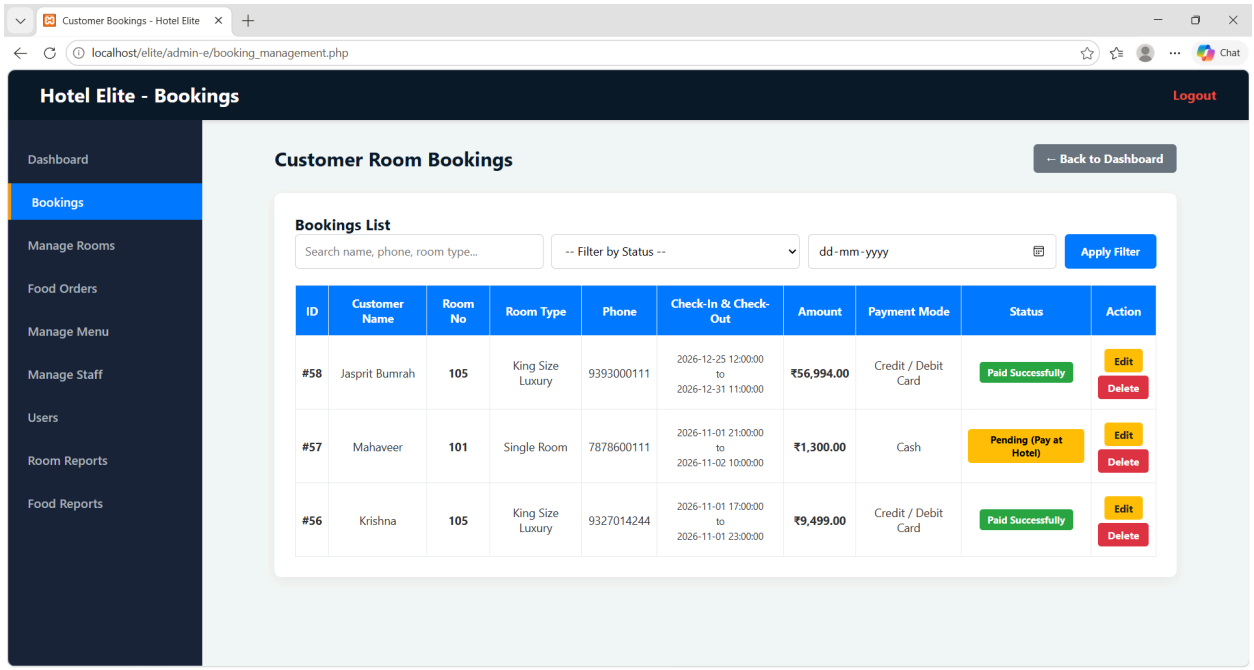
ADMIN PANEL
LOGIN PAGE



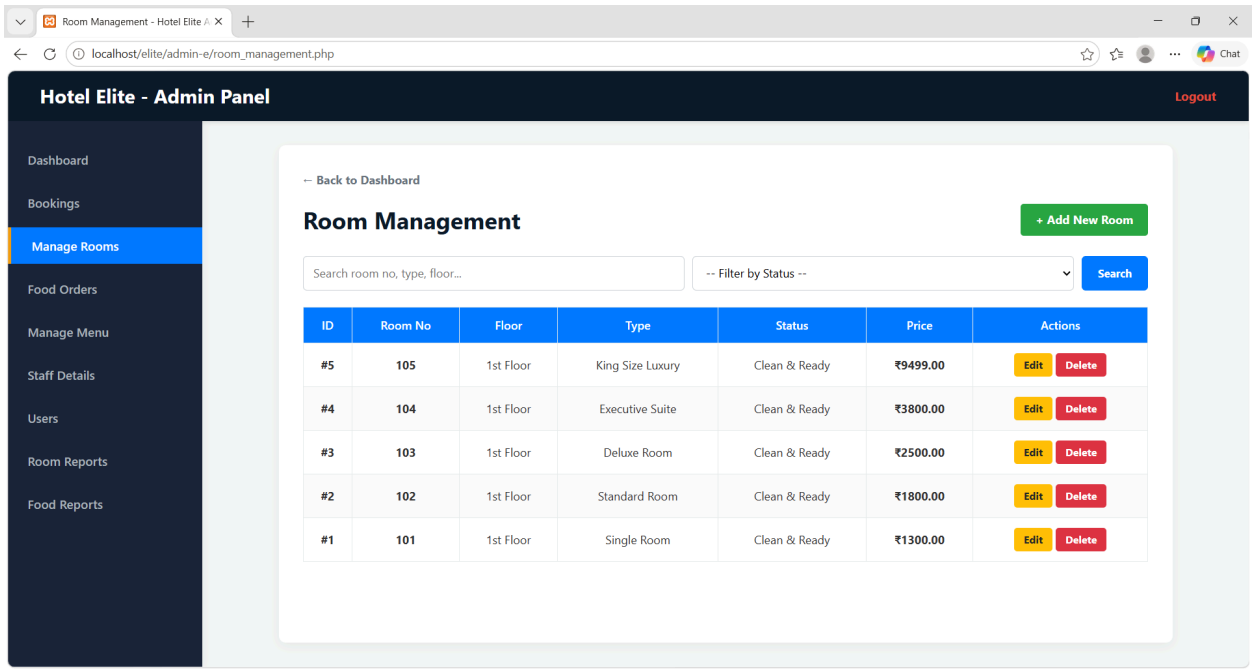
DASHBOARD



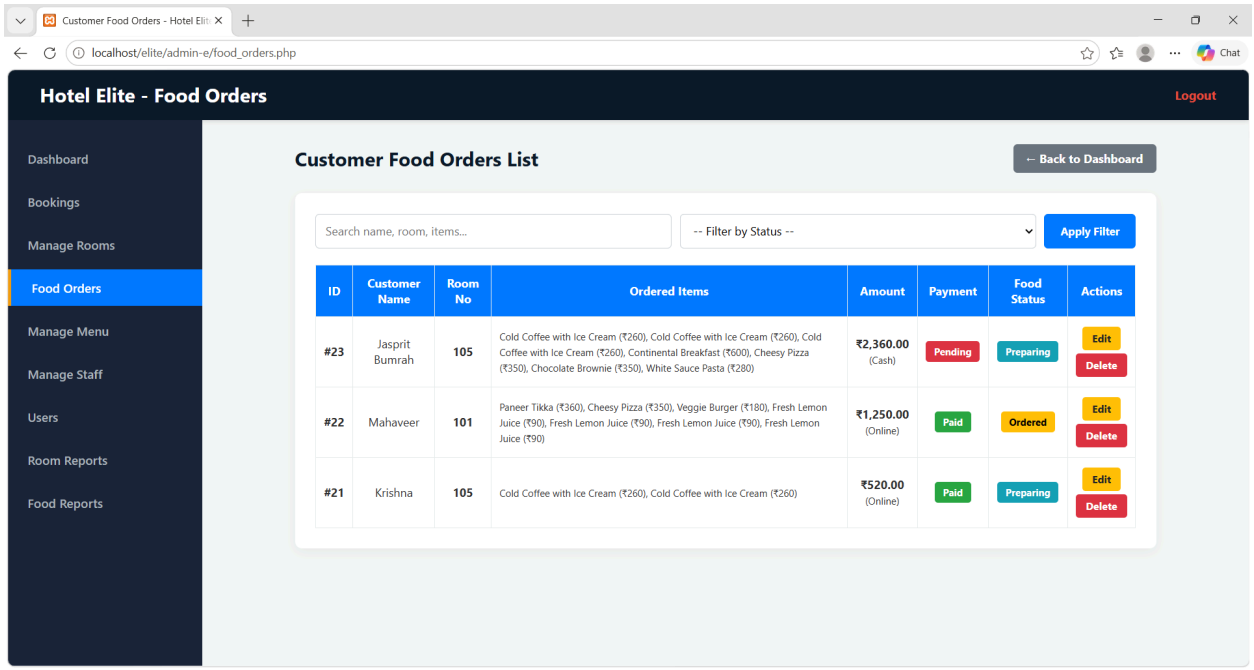
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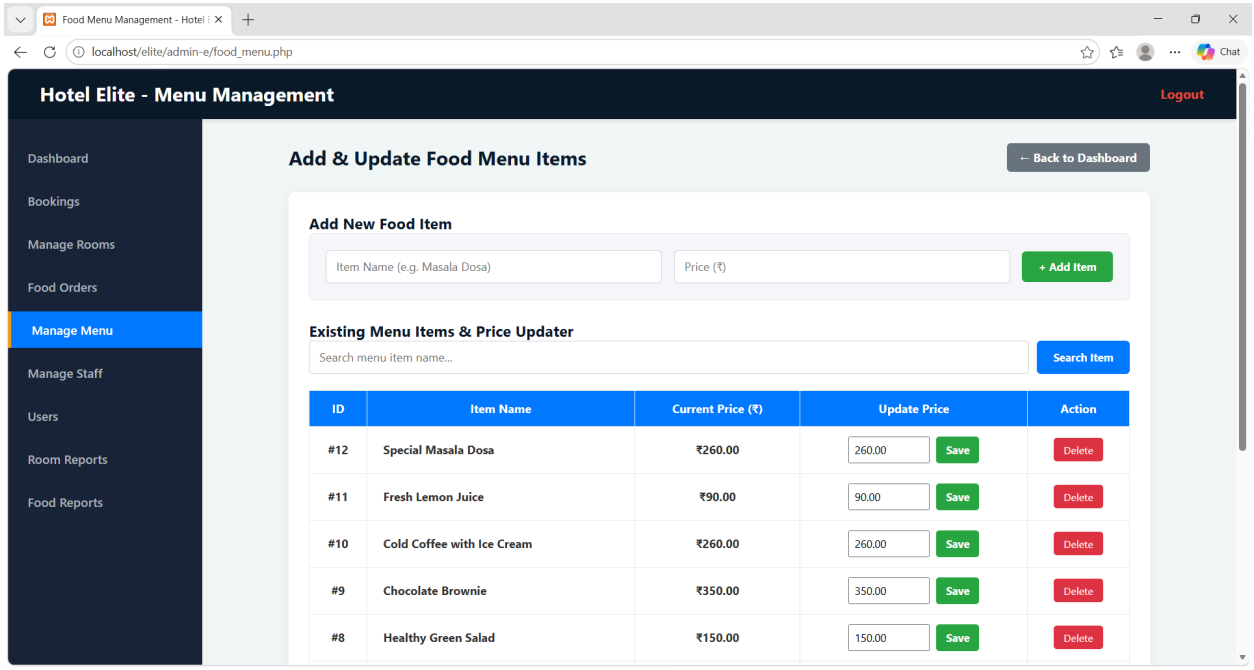
MANAGE ROOMS

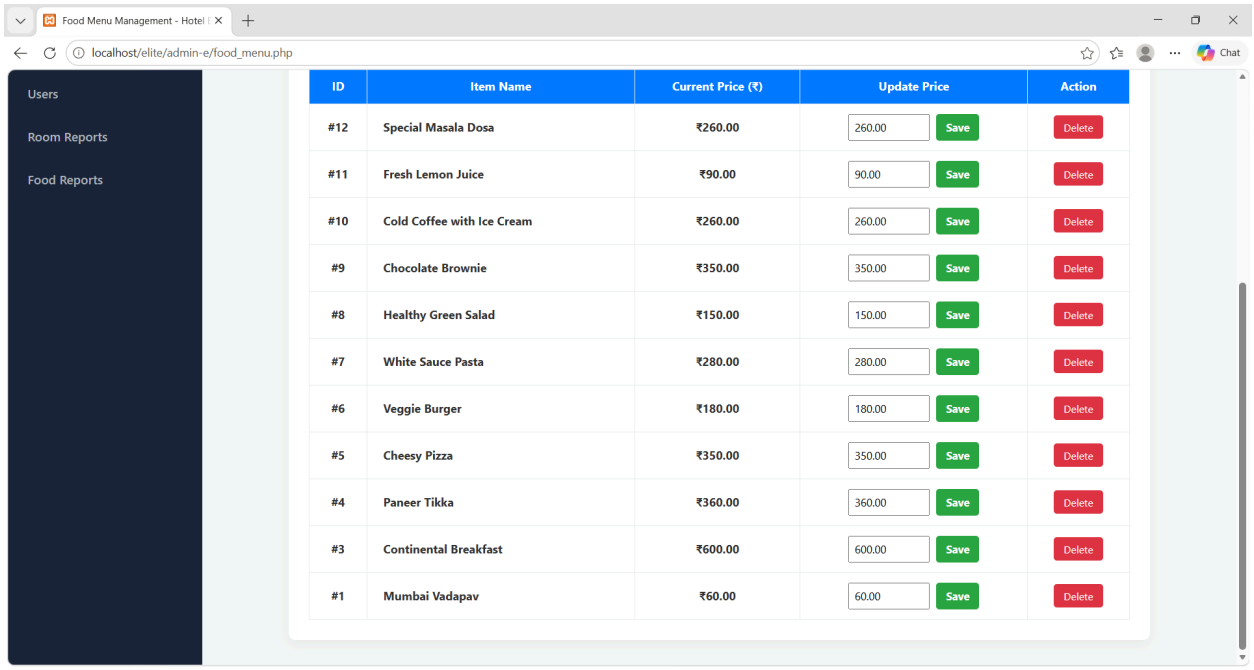


FOOD ORDERS

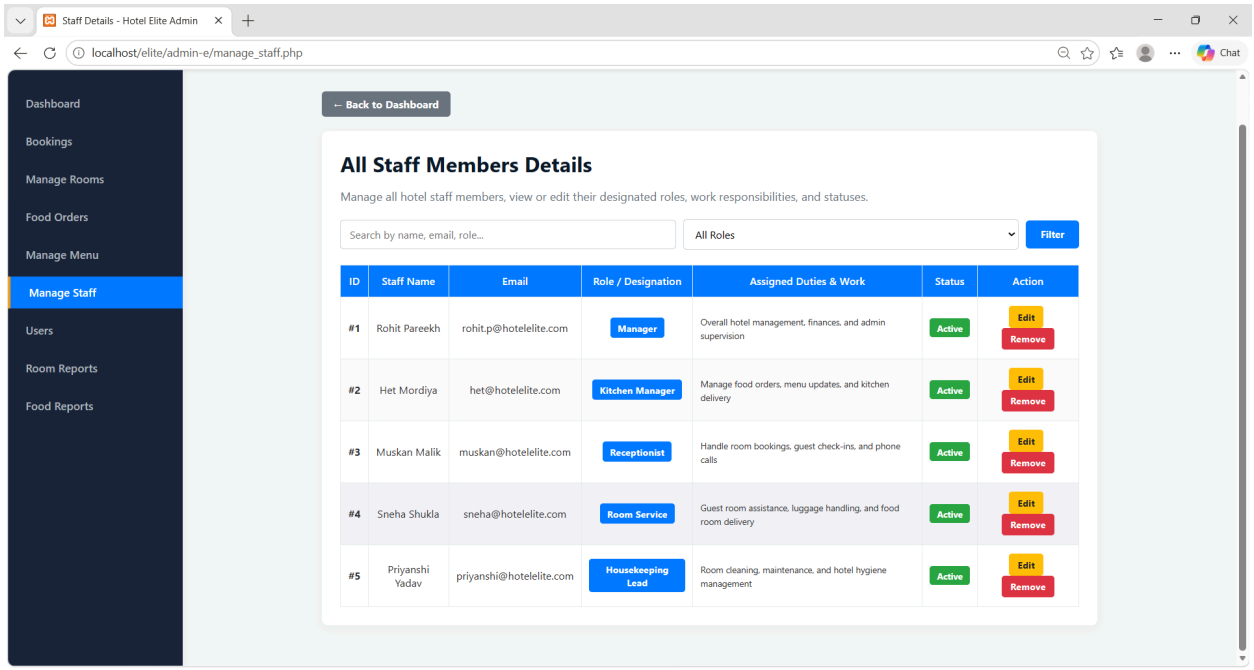


MANAGE MENU

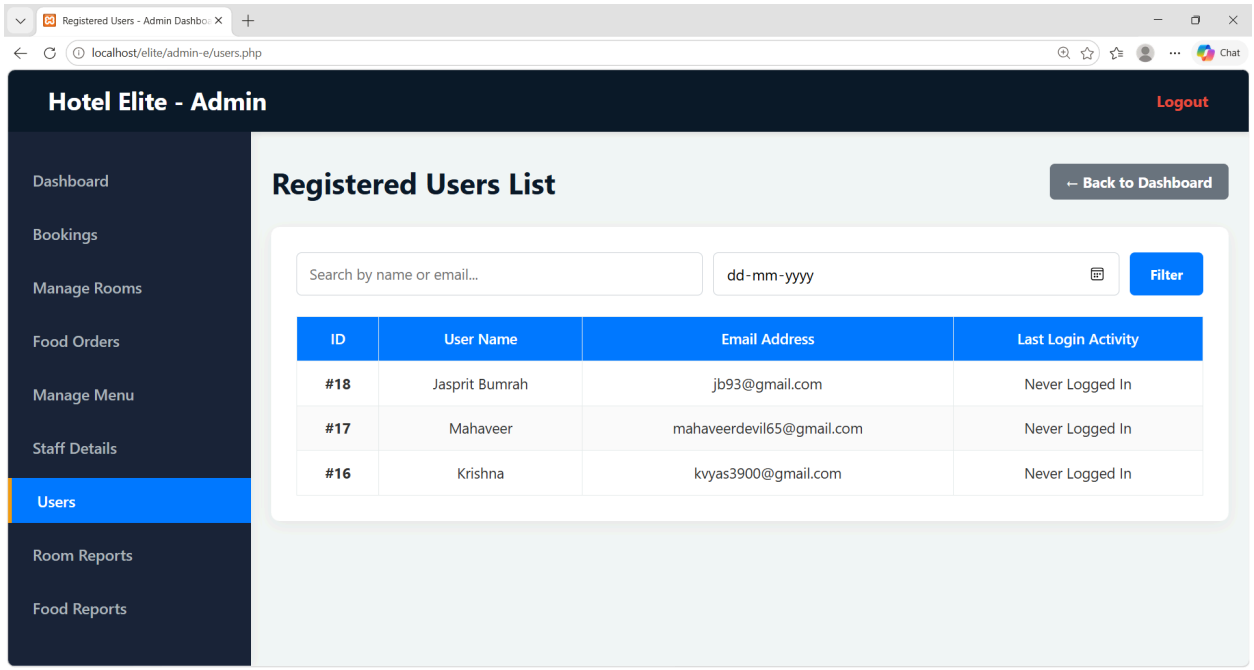




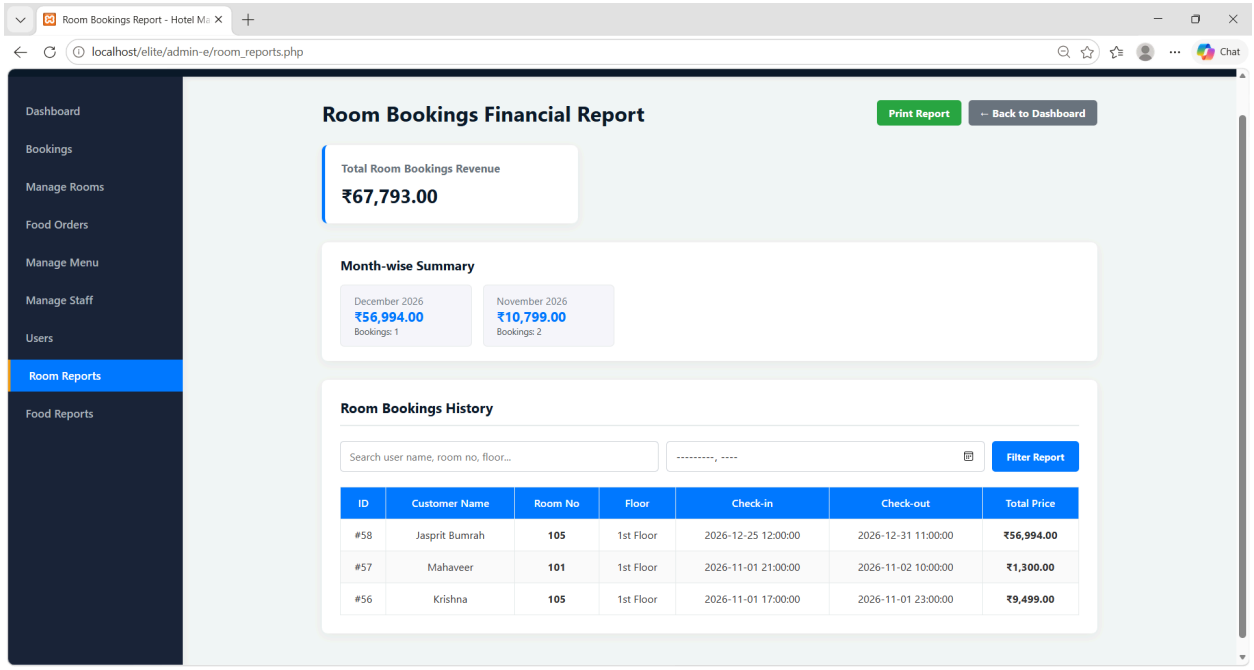
MANAGE STAFF



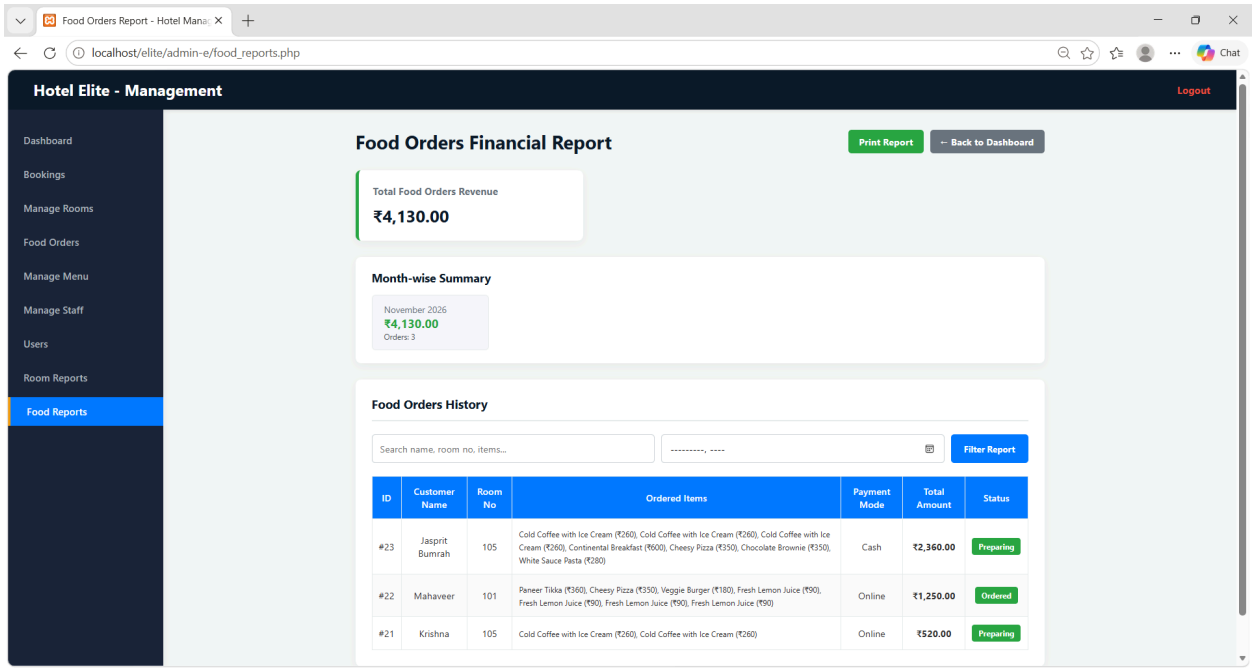
USERS



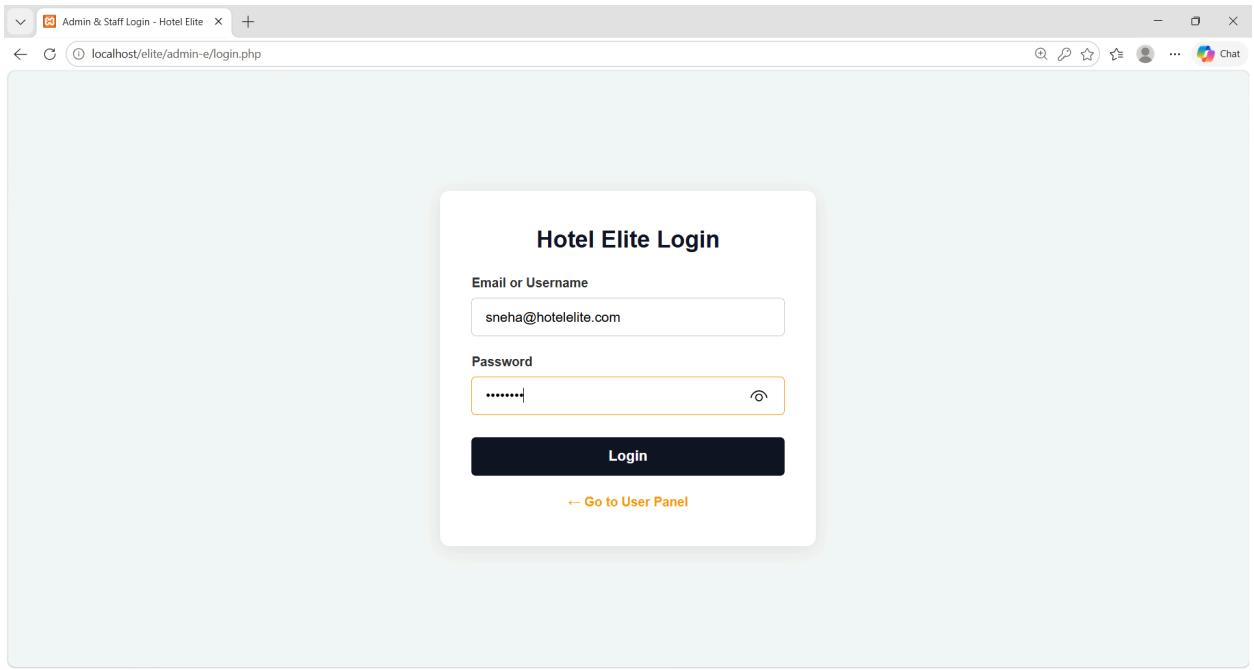
ROOM REPORTS



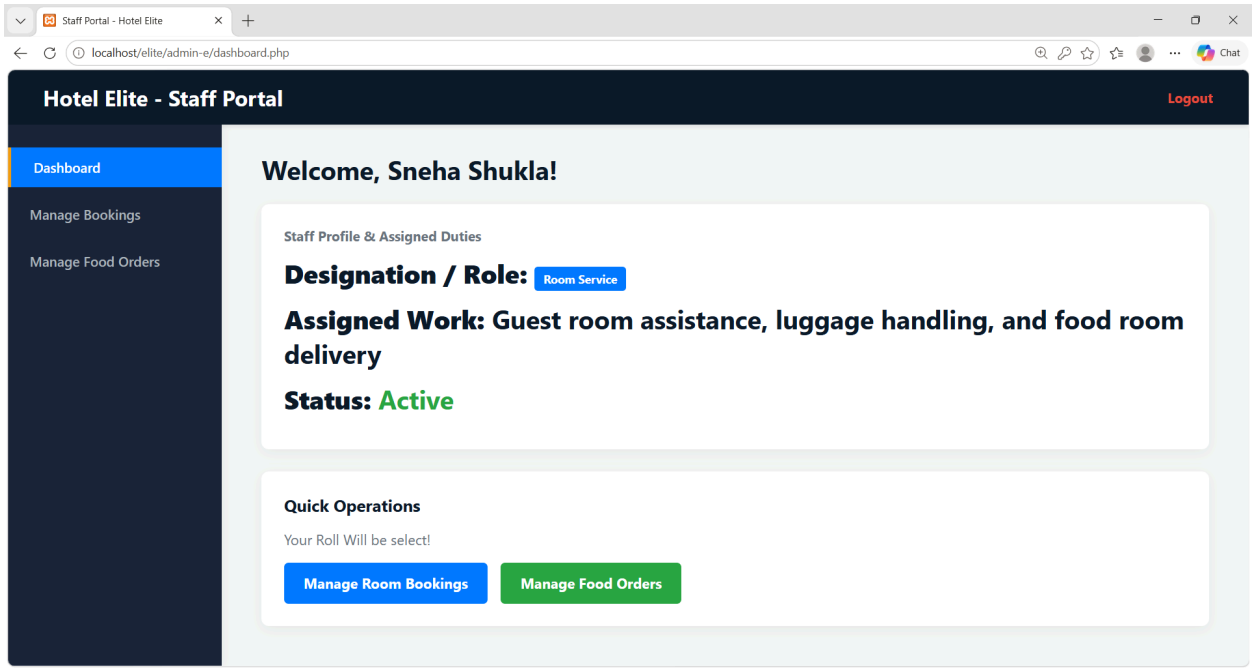
FOOD REPORTS



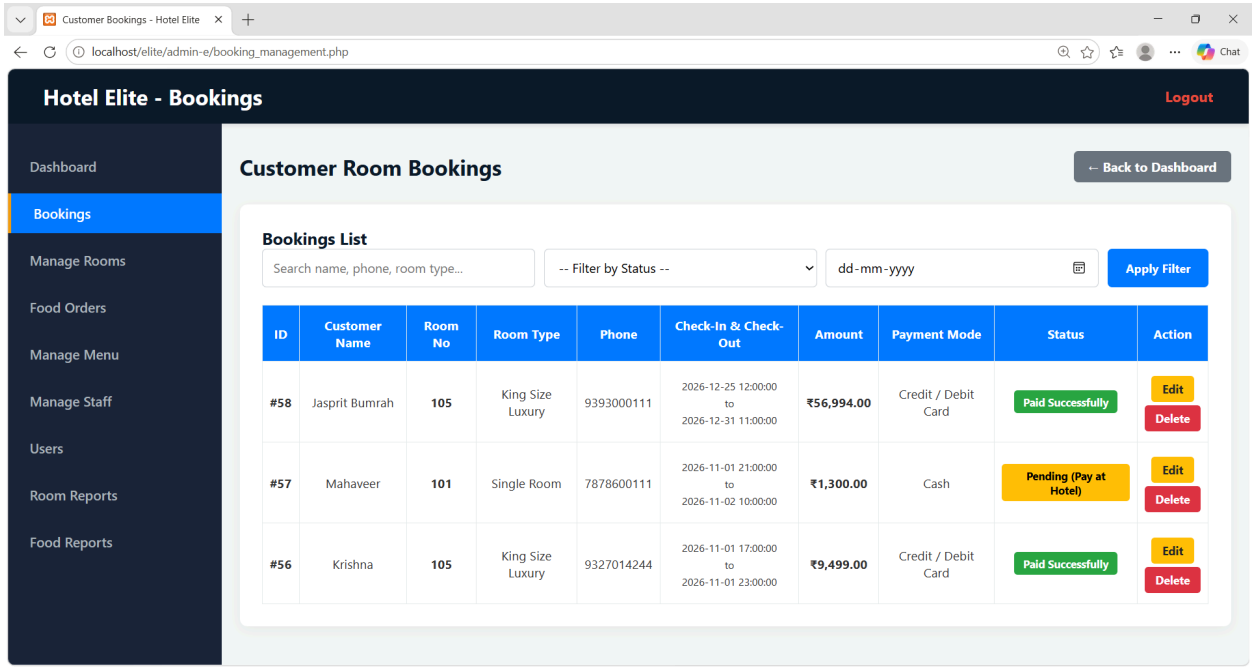
STAFF LOGIN



ROLL STAFF



MANAGE ROOM BOOKING



MANAGE FOOD ORDERS

